EXP 5: Creating Lambda Functions Using the AWS SDK for Python

```
CloudShell
eu-north-1 +
~ $ python3 -c "import boto3; print(boto3.__version__)"
1.42.1
~ $ nano trust-policy.json
~ $
```

```
CloudShell                                                                    Actions ▼
eu-north-1 +
  GNU nano 8.3                                  trust-policy.json                                  Modified
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "lambda.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was    M-F Next
```

```
~ $ aws iam create-role \
>  --role-name LambdaBasicRole \
> --assume-role-policy-document file://trust-policy.json
{
    "Role": {
        "Path": "/",
        "RoleName": "LambdaBasicRole",
        "RoleId": "AROA34AMC4VGCEPLPHOAL",
        "Arn": "arn:aws:iam::816069141836:role/LambdaBasicRole",
        "CreateDate": "2026-04-15T06:50:03+00:00",
        "AssumeRolePolicyDocument": {
            "Version": "2012-10-17",
            "Statement": [
                {
                    "Effect": "Allow",
                    "Principal": {
                        "Service": "lambda.amazonaws.com"
```

```
~ $ aws iam attach-role-policy   --role-name LambdaBasicRole --policy-arn arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
~ $ nano lambda_function.py
~ $
```

```
CloudShell                                                                    Actions ▼
eu-north-1 +
  GNU nano 8.3                                  lambda_function.py                                 Modified
def lambda_handler(event, context):
    return {
        "statusCode": 200,
        "body": "Hello from Lambda created using boto3!"
    }
```

```
~ $ zip function.zip lambda_function.py
  adding: lambda_function.py (deflated 17%)
~ $ nano create_lambda.py
~ $
```

```
  GNU nano 8.3                                                    create_lambda.py
import boto3

lambda_client = boto3.client("lambda", region_name="ap-south-1")

with open("function.zip", "rb") as f:
    zipped_code = f.read()

response = lambda_client.create_function(
    FunctionName="Boto3CreatedLambda",
    Runtime="python3.10",
    Role="arn:aws:iam::ACCOUNT_ID:role/LambdaBasicRole",
    Handler="lambda_function.lambda_handler",
    Code={"ZipFile": zipped_code},
    Description="Lambda created using boto3",
    Timeout=10,
    MemorySize=128,
    Publish=True
)

print(response["FunctionArn"])
```

```
~ $ python3 create_lambda.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue rece
iving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/develo
per/python-support-policy-updates-for-aws-sdks-and-tools/
  warnings.warn(warning, PythonDeprecationWarning)
arn:aws:lambda:eu-north-1:816069141836:function:Boto3CreatedLambda
~ $
```

```
CloudShell                                                                    Actions ▼
eu-north-1  +
  GNU nano 8.3                                                    invoke_lambda.py
import boto3
import json

client = boto3.client("lambda", region_name="ap-south-1")

response = client.invoke(
    FunctionName="Boto3CreatedLambda",
    InvocationType="RequestResponse",
    Payload=json.dumps({})
)

print(response["Payload"].read().decode())
```

```
~ $ nano invoke_lambda.py
~ $ python3 invoke_lambda.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue rece
iving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/develo
per/python-support-policy-updates-for-aws-sdks-and-tools/
  warnings.warn(warning, PythonDeprecationWarning)
{"statusCode": 200, "body": "Hello from Lambda created using boto3!"}
~ $
```

```
CloudShell                                                                    Actions ▼
eu-north-1  +
~ $ aws lambda delete-function --function-name Boto3CreatedLambda
{
    "StatusCode": 204
}
~ $ aws iam detach-role-policy \
>   --role-name LambdaBasicRole \
>   --policy-arn arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
~ $ aws iam delete-role --role-name LambdaBasicRole
~ $ ls
create_lambda.py  function.zip  invoke_lambda.py  lambda_function.py  trust-policy.json
~ $ rm *
~ $ ls
~ $
```